Data Analytics Project: Retail Sales Performance Analysis

Project Title

Retail Sales Analytics and Business Performance Optimization

Background Scenario

You are working as a Data Analyst for a retail company that sells products across multiple regions and customer segments.

Recently, management has noticed several business problems:

- Declining profits in certain regions
- Inconsistent customer purchasing behavior
- High sales in some products but low profitability
- Unclear understanding of product and customer performance

The management team wants data-driven insights and actionable recommendations to improve overall revenue and profitability.

Your task is to analyze the sales data and present meaningful business insights using Python and Power BI.

Dataset Requirement

Download the dataset from Kaggle.

https://www.kaggle.com/datasets

Recommended Dataset

Search in Kaggle for:

- **“Sample Superstore Sales”**
- OR **“Global Superstore Dataset”**

Project Objectives

Analyze the retail sales dataset and answer the following business questions:

1. Which product categories and sub-categories generate the highest profit?
2. Which regions or customers are underperforming?
3. How are discounts affecting profitability?
4. Which products should the company promote or discontinue?
5. How can the company improve revenue and customer retention?

Project Tasks

Task 1: Data Understanding and Cleaning

Perform complete data preprocessing using Python.

Requirements

- Identify missing values
- Handle incorrect data types
- Remove duplicate records
- Standardize column names if necessary
- Create useful new columns such as:
  - Profit Percentage
  - Order Month
  - Order Year
  - Discount Level
  - Customer Lifetime Value (optional)

Tools

Use:

- Python
- pandas

Task 2: Exploratory Data Analysis (EDA)

Perform detailed analysis using:

- pandas
- matplotlib
- seaborn

Required Analysis

A. Sales and Profit Analysis

Analyze sales and profit by:

- Category
- Sub-category
- Region
- State
- Segment

B. Time-Based Analysis

Analyze:

- Monthly sales trend
- Yearly profit trend
- Seasonal sales patterns

C. Product Performance

Find:

- Top 10 profitable products
- Bottom 10 loss-making products

D. Customer Analysis

Analyze:

- Top customers by revenue
- Customers with low profitability
- Repeat purchasing behavior

Task 3: Customer Segmentation Using RFM Analysis

Perform RFM Analysis:

RFM Components

- **Recency** → How recently the customer purchased
- **Frequency** → How often the customer purchases
- **Monetary Value** → How much money the customer spends

Segment customers into:

- High-value customers
- Regular customers
- At-risk customers
- Low-engagement customers

Required Output

- Customer segmentation table
- Visual representation of customer groups
- Business interpretation of each segment

Task 4: Business Problem Investigation

Go beyond visualization and investigate real business problems.

Analyze and explain:

1. Why do some categories have high sales but low profit?
2. Are discounts negatively impacting profit?
3. Which regions require business intervention?
4. Which products should receive more marketing focus?
5. Are there customers generating high sales but low profit?

Task 5: Visualization and Dashboard Development

Python Visualizations

Create meaningful charts using:

- matplotlib
- seaborn

Examples:

- Bar charts
- Line charts
- Heatmaps
- Scatter plots
- Distribution plots

Power BI Dashboard

Build an interactive dashboard containing:

Mandatory Dashboard Components

- Total Revenue KPI
- Total Profit KPI
- Profit Margin KPI
- Regional performance analysis
- Product/category performance
- Customer segmentation insights
- Monthly sales trend
- Discount vs Profit analysis

Dashboard Requirements

- Proper titles
- Filters/slicers
- Business-friendly design
- Interactive visuals

Task 6: Final Business Recommendations

Based on your analysis, provide actionable recommendations.

- What the company should stop doing
- Which products/categories should be promoted
- Which regions require improvement strategies
- Which customer segments should be targeted
- Pricing and discount strategy improvements
- Suggestions to improve profitability

Expected Deliverables

Students must submit:

1. Cleaned Dataset
2. Python Notebook (.ipynb)
3. Power BI Dashboard (.pbix)
4. Final Report or Presentation (PPT/PDF)

Important Requirement

Critical Thinking Requirement

Do NOT create charts without purpose.

For every visualization:

- Mention the business question being answered
- Explain the insight obtained
- Provide at least one recommendation based on the insight

Tools & Technologies

Use:

- Python
- pandas
- matplotlib
- seaborn
- Jupyter Notebook
- Power BI